

The empirical recurrence for OEIS A249851, derived from the entry's generating function

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Abstract

OEIS A249851 carries an empirical recurrence of order 12. The entry separately records a generating function, not as a conjecture but as a statement of fact, and the recurrence is a consequence of it: multiplying the generating function by the polynomial the recurrence supplies gives a POLYNOMIAL, and the recurrence then holds at every index past that polynomial's degree. No model of the objects being counted is needed, and no search: the whole argument is one polynomial division, carried out in exact arithmetic. The result is conditional on the recorded generating function, which is what the entry asserts and what is checked here against every published term.

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1 The two statements

OEIS A249851 is “Number of length $7+4 \cdot n$ arrays with no five consecutive terms having the maximum of any two terms equal to the minimum of the remaining three terms.”. It has offset 1 and begins

10, 3966, 223505, 4527441, 49270055, 354521087, 1901506446, 8194431990, ...

The entry states, as a conjecture and never marked settled:

$$a(n) = 12*a(n-1) - 66*a(n-2) + 220*a(n-3) - 495*a(n-4) + 792*a(n-5) - 924*a(n-6) + 792*a(n-7) - 495*a(n-8) + 220*a(n-9) - 66*a(n-10) + 12*a(n-11) - a(n-12) \text{ for } n > 12.$$

and, separately and NOT as a conjecture:

$$\text{G.f.: } x*(10 + 3846*x + 176573*x^2 + 2104937*x^3 + 8824523*x^4 + 14875683*x^5 + 10543155*x^6 + 3058279*x^7 + 321259*x^8 + 8535*x^9) / (1 - x)^{12}.$$

As of the “Last modified” line on the live entry (Aug 18 2017 18:11:20, revision 8) the first is still recorded as empirical. This note derives it from the second.

2 A rational generating function decides every linear recurrence

Write

$$G(x) = \sum_n a(n) x^n = \frac{8535x^{10} + 321259x^9 + 3058279x^8 + 10543155x^7 + 14875683x^6 + 8824523x^5 + 2104937x^4 + 8535x^3 + 321259x^2 + 176573x + 10}{x^{12} - 12x^{11} + 66x^{10} - 220x^9 + 495x^8 - 792x^7 + 924x^6 - 792x^5 + 495x^4 - 220x^3 + 66x^2 - 12x + 1}$$

the entry's own generating function, and let

$$D(x) = 1 - \sum_i c_i x^i = x^{12} - 12x^{11} + 66x^{10} - 220x^9 + 495x^8 - 792x^7 + 924x^6 - 792x^5 + 495x^4 - 220x^3 + 66x^2 - 12x + 1$$

be the polynomial attached to the conjectured recurrence.

Second, the polynomial division above was carried out symbolically, not numerically, so the statement that $D(x)G(x)$ is a polynomial is exact rather than approximate.

Third, the recurrence was evaluated directly on the series coefficients for sixty consecutive indices beyond $n = 10$ and holds at every one, and at $n = 10$ itself it fails — the degree bound is attained, so the range stated is the true one and not merely a safe one.

Remark 3. The argument is conditional on the generating function, which the entry records as a fact and attributes to a contributor rather than offering as a conjecture. What is shown here is that the empirical recurrence is not an independent guess: it is equivalent to information the entry already contains.

References

- [1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org>, 2026. Sequence A249851.
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- [3] M. Kauers and P. Paule, *The Concrete Tetrahedron*, Springer, 2011.
- [4] P. Flajolet and R. Sedgewick, *Analytic Combinatorics*, Cambridge University Press, 2009.